Understanding and verifying the autoregressive conditioning hypothesis

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Summary

This project explores how language models behave when asked to pursue specific goals - a mechanism we use to turn world models into agents. Using chess as our testing ground, we'll investigate whether models follow theoretical predictions when responding to prompts like "*find the best move*". Using carefully prepared training data, we will compare their behavior against a mathematical model of the “autoregressive conditioning hypothesis”. If successful, this work could potentially offer a new framework to analyze AI behavior, from a more abstract, implementation-agnostic point of view.

The goal

LLMs can be thought to be conditioned on goals by means of describing them, textually. By writing “`The best move in the position X is `”, we might be aiming to condition the model on winning - to then repeatedly, autoregressively sample those moves. This posterior is a very elegant and concise way to describe the resulting policy.

The question is: *is this what actually happens?* An affirmative answer constitutes the **autoregressive conditioning hypothesis**.

Our goal here will be answering that question empirically, in the context of chess (LLMs are notoriously weird about chess, and in the process, we might gain some understanding on why it is so). Investigating it properly requires having tight control over the training distribution - simplest to do when training models from scratch, but there are potentially other options to explore as well. Once we have that, we can use the prior formed by the dataset to compute the posteriors predicted by the autoregressive conditioning hypothesis analytically, compare those to the empirically observed moves played by the trained models, and check whether they match. **This is the Minimum Viable Product.**

Once we have this ready, we can aim for potential extensions and side questions:

- Most importantly, I would like us to iteratively train the models on their own outputs, and check whether their training trajectories match the theoretical predictions. If those agreed, it would improve our understanding on how exactly RLHF-/RLAIF-like methods might be modifying the base policies.
- Our prior theoretical work suggests that, if the autoregressive conditioning hypothesis is true, models will be susceptible to incoherence, which stems from the structure of the conditioning process, and distorts the posterior policy. Confirming this fact experimentally would be another good milestone.

- Having the training and data pipelines ready, by curating the dataset in a certain way (e.g. opponents always playing certain probability distributions), we could try to check whether models update on evidence correctly. More than that: we could check what their belief update function is, and whether it looks like the Jeffrey's update rule, or the Pearl's update rule, or potentially something else entirely.

Background

We will study the conditioning mechanism, through which an autoregressive predictive model is turned into an agent acting in an environment. Specifically: having an (abstract) world model, that is, a probability distribution $\pi(\tau)$ over all future world trajectories τ, one can turn it into an agent by constructing a policy $\pi(\tau|G)$ conditioned on a particular goal G. One then can sample an action $a \sim \pi(\tau|G)$, obtain a new state $s \sim \pi(s|\tau, a)$, and repeat the process in an auto-regressive fashion, $\tau' := (\tau, a, s)$. This is one model of how agentic behaviour can be elicited from a world-model-like system: in reinforcement learning literature, this approach is known as control-as-inference. In other contexts, it is also called active inference.

Note that so far we are not assuming any specific representation of the world model, any representation of the goal, or any implementation of the conditioning process - the abstract model can describe situations which were not designed to work this way *operationally*. This is, I think, a good reason to study it: we can potentially make progress on the alignment question by being somewhat implementation-agnostic. The intuition is that policies of the form $\pi(\tau|G)$ are simple and amenable to (safety) analysis, while real-world, large transformer models are complicated and messy. But if we knew that the latter approximate the former, we have an angle to reduce the complexity, and potentially prove, or at least analyse, their high-level properties.

Prior work

Myself and others have already done some preliminary work on understanding problems that goal-conditioning autoregressive generative models might create: in particular, what are (some) conditions under which this results in suboptimal policies. We also have a theory on how the policies and training trajectories of models look like, in an idealised scenario (unpublished WIP theory paper).

Project plan

Rough project plan:

1. First, we'll do a literature review of LLMs-and-chess, conditioning hypothesis, basic control-as-inference theory, and bring everyone up to speed.
2. Then, we'll re-implement a toy example of a simple sequential game (e.g. tic-tac-toe, connect-four, nim) to check we understand the problem.
3. The next task would be to find or create a good dataset (possibly starting from this prior work): we should preprocess it to make computing statistics relevant for estimating the posterior quick and simple.
4. After that, we'll brainstorm and implement a model architecture amenable to conditioning: the simplest approach would probably be a decision transformer

(although these have their issues, and in general can be expected to behave differently from LLMs trained on a more diverse corpus), or a clever way of encoding it in a Chess-GPT model. On the other hand, we could also try to train a nano-GPT model, and choose to focus the effort on dataset curation and filtration.
5. We will probably run into issues with comparing the distributions of the trained models with theoretical predictions - there are many possible measures, computing true conditional distributions is computationally hard, and some careful thinking and experimentation will be needed to see what works. Toy models developed in point 2 might help here.
6. Ideally, we will be writing up our results as we go - but at this point, we would consolidate it, preferably in the form of a (workshop) paper.
7. If time permits, we'll then investigate other questions (such as those I listed above).

This plan is flexible and will be adapted depending on the participants' tastes, how hard different parts turn out to be etc.

About myself

I am a third-year Computer Science PhD student at the University of Oxford. My research interests lie (mostly) at the intersection of mathematics and theoretical computer science - such as the semantics of programming languages, Markov chain Monte Carlo methods, reinforcement learning theory and information geometry. I currently split my time between doing research on categorical semantics of probabilistic programming languages and working on AI safety. You can find my personal website here.

My involvement and availability

I can commit to spending 10 hours per week on this project, but I expect myself to work more on average (although with high variance). I am probably going to be quite involved and hands-on, such as writing and reviewing code, debugging problems etc.

The team

I would like to work with a small team of 2-4, ideally consisting of people who have some mixed mathematical and computer science background. We will probably need to work quite intensely on the experimental side, although there should be opportunities for more theoretical work once we have our experimental setup ready. Availability and experience in training transformer models - including managing, cleaning and curating datasets - are a big advantage, but not strictly necessary. The baseline level of involvement would be around 10 hours per week.

On the experimental side, we will work with the standard ML toolkit (Python, PyTorch or JAX, HuggingFace etc.). The theory part is formulated in the language of Markov Decision Processes. You can skim the papers referenced in the **Prior work** section above to get a taste of it - even though our project will probably be less theory-focused.

How does this project advance AI safety?

There is a non-negligible chance that the current trend of deriving agents from unsupervised-trained world models is going to continue. On the other hand, specific implementation techniques, training setups and model architectures might very well change. Excluding cases of either extremely short or extremely long timelines, some middle ground in the level of abstraction is needed: control-as-inference approach is one I found to be the most convincing - the best model we have of the existing intelligence, humans, is to some extent founded on a similar theory.

Deep understanding of policies derived in this way could then shed light on what sort of failure modes we might expect from them, and how to counterbalance those. For example: survivorship bias present in the training data might make the resulting agents prone to self-preservation. Or: naively conditioning on results might lead to policies that pursue undesirable instrumental goals, resulting in power-seeking or incorrigibility.

Abstractions are certainly going to be leaky. Real models will probably deviate a lot from the idealised abstractions - due to the random training process dynamics and initialisation, insufficient data, inductive biases of the specific architecture etc. Delineating those boundaries and understanding where we can safely assume abstractions to hold - and where to break - would be the next step of the big-picture plan in this direction.